

ESTROGEN THERAPY AND THE RISK OF BREAST, OVARIAN, AND ENDOMETRIAL CANCER

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Introduction

Millions of women have used estrogens for relief of menopausal symptoms. Theoretical considerations, clinical observations, and epidemiologic information raise concern as to possible associations between use of exogenous estrogens and cancers of the female reproductive tract (Thomas, 1978). Because of the large number of women who have been using or will use estrogens for menopausal symptoms, any risk of or protection from these cancers could have major health impact on women in their postmenopausal years.

Convincing epidemiologic and clinical evidence indicates that estrogen therapy is a risk factor for the development of endometrial cancer (Hulka, 1980). Available studies do not indicate that estrogen therapy is a major risk factor for breast cancer, although there is some evidence that the risk may be increased in women who have received higher-dose or long-term estrogen therapy (Thomas, 1984). Little epidemiologic evidence exists concerning the association of estrogen therapy with ovarian cancer.

In this chapter, we summarize the available epidemiologic information

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concerning the relationship between estrogen therapy and cancer of the breast, ovary, and endometrium. In addition, we present preliminary information from the Cancer and Steroid Hormone Study, a multicenter, case-control study of breast, ovarian, and endometrial cancer. In general, our analyses support the results from other studies. However, we still do not know whether postmenopausal estrogen therapy is a risk factor for breast and ovarian cancer. Our evidence and the evidence of others suggest that it may be; clearly, these questions need to be carefully examined through studies especially designed for this purpose.

The Cancer and Steroid Hormone Study

The Cancer and Steroid Hormone Study is a large, case-control study coordinated by the Division of Reproductive Health at the Centers for Disease Control, with support from the National Institute of Child Health and Human Development and the National Cancer Institute (Centers for Disease Control Cancer and Steroid Hormone Study, 1983a,b,c). Although the study was designed primarily to examine the relationship between the use of oral contraceptives (OCs) and breast, endometrial, and ovarian cancer, information was also collected on noncontraceptive estrogen use. Study participants were enrolled between December 1980 and April 1983 from the geographic areas of eight regional tumor registries of the Surveillance, Epidemiology and End Results Centers of the National Cancer Institute. These registries were located in the urban areas of Atlanta, Detroit, Seattle, and San Francisco, and the states of Utah, Iowa, Connecticut and New Mexico.

The study methods have been described in detail elsewhere (Centers for Disease Control Cancer and Steroid Hormone Study, 1983a,b,c). Briefly, a case was defined as a woman aged 20–54 years with histologically confirmed breast, endometrial, or ovarian cancer newly diagnosed during the study period who resided in one of the eight areas at the time of diagnosis. The control group consisted of women selected by randomly telephoning households in the same geographic areas as the cases to provide a group of women 20–54 years of age reflecting the general population of these areas (Waksberg, 1978). These women were frequency-matched by 5-year age group to the expected age distribution of the women with breast cancer.

All cases and controls were administered an hour-long questionnaire concerning their reproductive, medical and family histories. A calendar marking important life events was also used to aid the women's recall of their use of OCs and other female hormones. To help them remember specific brands and doses of the OCs and exogenous estrogens they used, the women were shown a book of photographs of the major brands marketed in the United States.

Breast Cancer

This year 37,000 women will die of breast cancer in the United States: 75% of these deaths will occur among women over the age of 54 (Silverberg and Lubera, 1983). If estrogen therapy increases a woman's risk of breast cancer, this could have important public health impact. Despite a number of epidemiologic studies, the relationship between use of estrogens and subsequent breast cancer risk is not clear.

One important retrospective cohort study examined the relationship of estrogen therapy to breast cancer risk by comparing the observed number of breast cancer cases in the study cohort to the expected number in the general population, derived from national survey data (Hoover *et al.*, 1976). The study cohort was a group of women in one physician's practice who had received conjugated estrogen therapy during the period from 1939 to 1972. During this time, 49 women developed breast cancer, whereas 39 would have been expected from the national data, resulting in a risk of 1.3 of developing breast cancer in the cohort of women, compared to the general population of women. This finding was not statistically significant. However, the risk was 2.0, and statistically significant, in women who had been followed in the study for at least 15 years.

At least 12 case-control studies of breast cancer and estrogen therapy have been published since 1974 (Boston Collaborative Drug Surveillance Program, 1974; Craig *et al.*, 1974; Casagrande *et al.*, 1976; Sartwell *et al.*, 1977; Rav-nihar *et al.*, 1979; Jick *et al.*, 1980; Ross *et al.*, 1980; Brinton *et al.*, 1981; Hoover *et al.*, 1981; Kelsey *et al.*, 1981; Hulka *et al.*, 1982; Kaufman *et al.*, 1984). The studies vary in methodology and in size, but in almost all, the relative risk of breast cancer in women who had ever used estrogens was approximately 1 compared with the risk for women who had never used estrogens, suggesting little overall association between estrogen use and breast cancer.

Table I presents information from published studies concerning the risk of breast cancer, considering the duration or dosage of estrogen therapy. No trend in risk by duration of use is apparent; however, Hoover found a steady trend of increasing risk of breast cancer with increasing number of estrogen prescriptions filled (Hoover *et al.*, 1981). Likewise, some studies show increasing risks with increasing daily dose or increasing cumulative dose of estrogen (Ross *et al.*, 1980; Brinton *et al.*, 1981; Hoover *et al.*, 1981). These studies suggest that higher-dose estrogen therapy may increase a woman's risk of breast cancer by up to twofold. However, other studies have not found evidence for a dose-response relationship (Kelsey *et al.*, 1981; Hulka *et al.*, 1982; Kaufman *et al.*, 1984).

Additional evidence that high dosages of estrogens can cause breast cancer comes from a report that two transvestite men who received massive doses of

Table I
Published Risk Estimates of Breast Cancer in Relation to Various Doses of Estrogens^a

Measurement of dose	Study	Dose category	Relative risk ^b
Duration of use	Ravnihar <i>et al.</i> (1979)	< 2 years	0.7
		≥ 2 years	0.7
	Ross <i>et al.</i> (1980)	≥ 7 years	1.8
	Brinton <i>et al.</i> (1981)	< 5 years	1.2
		5-9 years	1.3
		≥ 10 years	1.2
	Hulka <i>et al.</i> (1982) ^c	≤ 3 years	2.1
		4-9 years	1.5
		≥ 10 years	1.7
	Kaufman <i>et al.</i> (1984) ^c	< 1 year	0.9
		1-4 years	0.9
		5-9 years	0.7
		≥ 10 years	1.3
Number of prescriptions	Hoover <i>et al.</i> (1981)	1	1.0
		2-4	1.3
		5-9	1.5
		≥ 10	1.7
Daily dose	Hoover <i>et al.</i> (1981)	< 1.25 mg	1.1
		≥ 1.25 mg	1.5
	Brinton <i>et al.</i> (1981)	0.3 mg	0.8
		0.6 mg	1.3
		1.25 mg	1.2
		2.5 mg	1.7
	Hoover <i>et al.</i> (1976)	0.3 mg	1.2
		0.625 mg	1.1
		> 0.625 mg	1.9
	Hulka <i>et al.</i> (1982) ^c	≤ 0.625 mg	1.9
		> 0.625 mg	1.0
	Kaufman <i>et al.</i> (1984) ^c	< 1.25 mg	1.2
		≥ 1.25 mg	0.7
Cumulative dose	Ross <i>et al.</i> (1980)	1-1499 mg/life	0.8
		≥ 1500 mg/life	1.9
	Kelsey <i>et al.</i> (1981)	1-49 mg-months	0.8
		≥ 50 mg-months	0.6

^a Modified from Thomas (1984).

^b Risk among users relative to never-users.

^c Study limited to women with a natural menopause.

estrogens to induce breast development subsequently developed breast cancer (Symmers, 1968).

The estrogens used by women in these studies were predominantly conjugated estrogens; the findings thus reflect largely the influence of conjugated estrogens on the risk of breast cancer.

Since the Cancer and Steroid Hormone Study was designed primarily to examine OC use, we did not enroll cases or controls over the age of 54, because older women were unlikely to have ever used OCs. For the analysis of breast cancer and postmenopausal estrogens, we included only women who had had a surgical or a natural menopause. By definition, women with a surgical menopause had had a hysterectomy causing cessation of their menses; they may or may not have had intact ovaries. Additionally, if a woman had used estrogens before her last menstrual period, she was excluded from the analysis. After these exclusions we had data for analysis from 1352 women with breast cancer and 1617 controls. The results are derived from a logistic regression analysis that controlled for current age, number of ovaries, type of menopause, age at last menstrual period, age at first full-term pregnancy, history of benign breast disease, and family history of breast cancer.

As expected, compared with controls, the women with breast cancer were more likely to be nulliparous (14% vs. 9%), to have a history of benign breast disease (21% vs. 13%), and to have a family history of breast cancer (27% vs. 20%). About two thirds of both cases and controls had had a surgical menopause.

The highest relative risks of breast cancer were found in those women with the longest duration of estrogen use, but none of the risk estimates was significantly different from 1 (Table II). Note that the number of women using estrogens for 15 or more years was small, so that the risk estimates for these duration categories are unstable. Because the enrollment cutoff in the study was at age 54, most of the postmenopausal women who ever used estrogens used them after a surgical rather than a natural menopause. The results, therefore, largely reflect the risks among women having a surgical menopause. We examined the risk of breast cancer among only the women with a surgical menopause and

Table II
Risk of Breast Cancer by Duration of Postmenopausal Estrogen Use^a

Duration of Estrogen use ^b	Cases	Controls	Relative risk ^c (95% confidence interval)
Never	959	1135	1 (referent)
< 1 year	74	87	1.1 (0.8-1.5)
1-4 years	159	193	1.0 (0.8-1.3)
5-9 years	98	113	1.1 (0.8-1.6)
10-14 years	33	57	0.8 (0.5-1.3)
15-19 years	17	20	1.4 (0.7-2.8)
≥ 20 years	7	6	1.6 (0.5-5.0)

^aFrom Centers for Disease Control Cancer and Steroid Hormone Study.

^bExcludes five cases and six controls with unknown duration of use.

^cLogistic regression estimate, adjusted for age, number of ovaries, type of menopause, age at last menstrual period, age at first full-term pregnancy, history of benign breast disease, and family history of breast cancer.

found similar results; that is, the risk was still highest among the women with a surgical menopause who had used estrogens for 15 or more years. There were too few women with a natural menopause who had a long duration of estrogen use to determine the risks for this group.

We considered not only the duration of estrogen received but also the time that had elapsed since a woman first received postmenopausal estrogen therapy, called the latent period. Here, the highest risks were seen for women who had first used estrogens 15 or more years ago (Table III).

The risk of breast cancer associated with postmenopausal estrogen use was not altered by a woman's parity, her history of benign breast disease, or her family history of breast cancer. Neither was it altered by the presence or absence of ovaries.

This preliminary analysis has not resolved the controversy surrounding the possibility that postmenopausal estrogen therapy may increase a woman's risk of breast cancer. The study's primary limitation is that women over the age of 54 were not enrolled; hence, large numbers of postmenopausal women who may or may not have taken estrogens were not eligible for the study. Moreover, most women in the study who took estrogens for a long period of time had a surgical menopause, and estrogen's effects on these women may not be similar to its effects on women who had a natural menopause. Even with these limitations, this is still the largest study of the question, providing data on a substantial number of women with a surgical menopause who have used estrogens for at least 15 years. We have not found a large protective or large harmful effect. We have found a suggestion of a one and one-half to two-fold increased risk in those women who have received long-term estrogen therapy. This association, however, may be due to chance or an undetected bias.

Whether estrogen therapy increases a woman's risk of breast cancer needs

Table III
Risk of Breast Cancer by Time Since First Postmenopausal Estrogen Use^a

Time since first estrogen use ^b	Cases	Controls	Relative risk ^c (95% confidence interval)
Never	959	1135	1 (referent)
< 1 year	22	31	0.9 (0.5-1.6)
1-4 years	114	138	1.0 (0.8-1.4)
5-9 years	142	161	1.1 (0.9-1.5)
10-14 years	67	99	0.9 (0.7-1.3)
15-19 years	27	32	1.3 (0.7-2.2)
≥ 20 years	16	15	1.5 (0.7-3.2)

^aFrom Centers for Disease Control Cancer and Steroid Hormone Study.

^bExcludes five cases and six controls with an unknown time since first use.

^cLogistic regression estimate, adjusted for age, number of ovaries, type of menopause, age at last menstrual period, age at first full-term pregnancy, history of benign breast disease, and family history of breast cancer.

to be studied further through studies specifically designed to answer the question, with special consideration of estrogen's effects on women at high risk for breast cancer and on women with long duration of estrogen use.

Ovarian Cancer

Ovarian cancer is much less common than endometrial cancer or breast cancer; however, because of the higher case-fatality rate, ovarian cancer accounts for a disproportionately large number of cancer deaths each year. This year about 11,500 women will die of ovarian cancer.

Of the three cancers we have discussed, least is known about the relationship between estrogen therapy and ovarian cancer. In the late 1970s, before any epidemiologic evidence was available, Stadel proposed that use of estrogens might lower a woman's risk of ovarian cancer because estrogens lowered pituitary gonadotropins (Stadel, 1975). Since then, at least seven epidemiologic studies have been published that examine the relationship between estrogen use and ovarian cancer (Hoover *et al.*, 1977; Annegers *et al.*, 1979; McGowan *et al.*, 1979; Hildreth *et al.*, 1981; Weiss *et al.*, 1982; LaVecchia *et al.*, 1982; Cramer *et al.*, 1983). Six of the seven were case-control studies.

The one cohort study by Hoover (Hoover *et al.*, 1977) used the same cohort of women previously described in the breast cancer section. They followed up 908 women who had been treated with conjugated estrogens during the period from 1939 to 1969. During that time eight cases of ovarian cancer developed, whereas 3.4 cases would be expected, yielding a relative risk of 2.4 for this group of women compared to the general population of women. The researchers found no increased risk with increasing duration of use but did find an increased risk with increasing strength of estrogen tablet used. Of note, in the small group of women who had used diethylstilbestrol (DES) for 1 year or more in addition to conjugated estrogens, three women developed cancer whereas 0.1 case was expected. This gave a relative risk of 30, which was highly statistically significant.

Table IV summarizes results from the six case-control studies of ovarian cancer and estrogen therapy. Note that the number of ovarian cancer cases in each study was small. No statistically significant overall increased risk was found in any of the studies, although Cramer's study suggested that the risk may be higher in women with a long duration of use and in those women who had had a natural menopause (Cramer *et al.*, 1983). Annegers' study was the only one that found a protective effect, but this effect disappeared if women had used estrogens for more than six months (Annegers *et al.*, 1979). Regarding type of estrogen used, Weiss did not find an increased risk with DES use (Weiss *et al.*, 1982); Cramer found similar risks associated with conjugated and non-conjugated estrogens (Cramer *et al.*, 1983). Three of the studies examined the

Table IV
Risk Estimates from Case-Control Studies of Estrogen Therapy and Ovarian Cancer

Study	Age range	No. of cases	Ever use: relative risk ^a	Duration: relative risk ^a
Annegers <i>et al.</i> (1979)	≥ 15	116	0.5 ^b	≥ 6 months, 1.0
McGowan <i>et al.</i> (1979)	Average, 52	197	No association	Not stated
Hildreth <i>et al.</i> (1981)	45-74	62	0.9	No trend
Weiss <i>et al.</i> (1982)	50-74	207	1.3	No trend
LaVecchia <i>et al.</i> (1982)	40-69	135	1.0	No trend
Cramer <i>et al.</i> (1983)	Natural menopause	92	2.0	Not stated
	40-80	173	1.6	≤ 1 yr, 1.5 2-5 yr, 1.0 > 5 yr, 2.8

^aRisk relative to never-users.

^b*P* < 0.05.

association with estrogen use by histologic type of ovarian cancer; all found that estrogen therapy was a stronger risk factor for endometrioid ovarian cancer than for ovarian cancer of other types (Weiss *et al.*, 1982; LaVecchia *et al.*, 1982; Cramer *et al.*, 1983).

Overall, the results from these seven studies are inconclusive. However, the weight of evidence does not support the hypothesis that estrogen therapy protects a woman from ovarian cancer.

To examine the relationship between ovarian cancer and postmenopausal estrogen therapy using data from the Cancer and Steroid Hormone Study, we limited the study group to women who had either a natural or surgical menopause. However, because women who do not have ovaries are not at risk for developing ovarian cancer, women with no ovaries were also eliminated from the control group. Hence, all women in this analysis had at least one ovary. After these exclusions, we had data on 160 women with ovarian cancer and 1223 controls. A panel of three expert pathologists reviewed histology slides for over 90% of the ovarian cancer cases. We used logistic regression techniques to adjust the risk estimates for age, parity, OC use, and type of menopause.

Parity has an important association with ovarian cancer risk, so we were not surprised that women with ovarian cancer were more likely to be nulliparous or of low parity than were the controls. Controls (48%) were more likely to have had a surgical menopause than were cases (33%). These women did not have a surgical climacteric, since we included in the ovarian cancer analysis only women with at least one ovary.

Women who had ever used postmenopausal estrogens had a risk of 1.3 of developing ovarian cancer as compared with women who had never used post-

Table V
Risk of Ovarian Cancer by Duration of Postmenopausal Estrogen Use^a

Duration of estrogen use ^b	Cases	Controls	Relative risk ^c (95% confidence interval)
Never	134	1007	1 (referent)
Ever	26	203	1.3 (0.8-2.1)
<2 years	9	83	1.1 (0.5-2.3)
2-5 years	8	63	1.3 (0.6-2.8)
≥6 years	9	57	1.7 (0.8-3.6)

^aFrom Centers for Disease Control Cancer and Steroid Hormone Study.

^bExcludes four controls with unknown duration of use.

^cLogistic regression estimate, adjusted for age, parity, type of menopause, and history of oral contraceptive use.

menopausal estrogens, although this difference was not statistically significant (Table V). There was a trend of increasing risk with increasing duration of estrogen use, with a relative risk of 1.7 for women who had used estrogens for more than 6 years as compared with women who had never used estrogens.

When we examined the risk of ovarian cancer by time since first use of estrogens — the latent period — we found that compared with women who had never used estrogens, women who had first used estrogens more than 10 years ago had a risk of ovarian cancer of 2.0 (95% confidence interval: 1.0-4.1), while women who had first used estrogens more recently had a risk of 1.0 (0.6-1.8). The effect of increased latency was apparent regardless of whether a woman had used estrogens for a short or a long period of time.

Because OC use is known to protect a woman from developing ovarian cancer, we wondered if a past history of OC use might somehow modify the association of postmenopausal estrogen therapy and ovarian cancer. We computed the risk estimates for women who had and had not used postmenopausal estrogens by their past history of OC use (Table VI). Compared with women who had never used estrogens or OCs, those who had ever used both estrogens and OCs had an overall relative risk of 0.7 of developing ovarian cancer, whereas women who had used estrogens but had not used OCs had an overall risk of 1.7. Neither of these were statistically significant. When we examined the risks by duration of estrogen use we found that among the women who had never used OCs, those who had used estrogens for 6 or more years had a risk of 3.1 as compared with women who had never used estrogens. This was a statistically significant increased risk. Of note, in this group there were eight cases and 24 controls.

Almost all of the women in the study who had postmenopausal estrogen therapy had used conjugated estrogens. We had no case and only one control who had ever used DES. Hence, we could not analyze our data by type of estrogen used.

Because three previous studies suggested that the endometrioid type of

Table VI
Risk of Ovarian Cancer by Postmenopausal Estrogen Use and by Use of Oral Contraceptives^a

Duration of estrogen use ^b	Use of oral contraceptives			
	Never		Ever	
	Cases/controls	Relative risk ^c	Cases/controls	Relative risk ^c
Never	87/502	1 (referent)	47/505	0.7
Ever	18/86	1.7	8/117	0.6
< 2 years	6/34	1.4	3/49	0.5
2-5 years	4/28	1.1	4/35	1.0
≥ 6 years	8/24	3.1 ^d	1/33	0.2

^aFrom Centers for Disease Control Cancer and Steroid Hormone Study.

^bExcludes 4 controls with unknown duration of estrogen use.

^cLogistic regression relative risk, adjusted for age, parity, and type of menopause.

^d $P < 0.05$.

ovarian cancer had a higher risk associated with estrogen therapy than did the other types of ovarian cancer (Weiss *et al.*, 1982; LaVecchia *et al.*, 1982; Cramer *et al.*, 1983), we examined our data by histologic type of ovarian cancer. Although the numbers of women studied were small, we found no difference in risk for the different histologic types.

In this analysis, we are limited by the small number of cases of ovarian cancer and by the enrollment cutoff at age 54. Our findings are consistent with those of Cramer (Cramer *et al.*, 1983), which suggest that there may be an increased risk of ovarian cancer with long duration of estrogen therapy. However, our findings are also consistent with no effect of estrogen therapy on the risk of ovarian cancer.

These eight epidemiologic studies indicate that estrogen therapy neither protects against ovarian cancer nor has a large harmful effect. If an increased risk does exist, OC use may minimize the risk. But no study to date has been able to satisfactorily examine the effects of long duration and long latency of estrogen therapy on the risk of ovarian cancer. Resolution of these questions will require a study with a much larger number of ovarian cancer cases in the postmenopausal age group.

Endometrial Cancer

Much evidence exists concerning the relationship between estrogen therapy and endometrial cancer. Fortunately, mortality rates are low from this cancer; still, 39,000 new cases are diagnosed in the United States each year.

Epidemiologic evidence from at least 14 case-control studies published since the 1970s has consistently shown a positive relationship between estrogen therapy and adenocarcinoma of the endometrium, by far the most common type of endometrial cancer (Table VII) (Smith *et al.*, 1975; Ziel and Finkel, 1975; Mack *et al.*, 1976; Gray *et al.*, 1977; McDonald *et al.*, 1977; Horwitz and Feinstein, 1978; Hoogerland *et al.*, 1978; Antunes *et al.*, 1979; Hulka *et al.*, 1980; Shapiro *et al.*, 1980; Jelovsek *et al.*, 1980; Spengler *et al.*, 1981; Stavvaky *et al.*, 1981; Kelsey *et al.*, 1982). Each one of these studies has found that estrogen therapy, especially long-term therapy, is a risk factor for developing endometrial cancer. The relative risks for women who have ever used estrogens compared to never-users have ranged from 2 to 12, depending on such factors as the control group used and the durations considered. For long-term estrogen therapy, the risks have ranged from 4 to 15. Most of these studies have examined risks associated with use of conjugated estrogens, but there is no reported evidence of differences in risk associated with different types of estrogens.

Convincing evidence indicates that adding progestin therapy to estrogen therapy prevents the development of hyperplastic changes in the endometrium, considered the precursor to adenocarcinoma (Paterson *et al.*, 1980). Although the hypothesis is biologically plausible, sufficient epidemiologic evidence has not yet been published to determine whether combined estrogen-progestin therapy actually reduces the risk of endometrial cancer.

Table VII
Risk Estimates from Case-Control Studies of Estrogen Therapy
and Endometrial Cancer

Study	Relative risks ^a	
	Ever	Long-term
Smith <i>et al.</i> (1975)	4.5	—
Ziel and Finkel (1975)	7.6	13.9
Mack <i>et al.</i> (1976)	5.6	8.8
Gray <i>et al.</i> (1977)	3.1	11.6
McDonald <i>et al.</i> (1977)	2.0	7.9
Horwitz and Feinstein (1978)	12.0	—
Hoogerland <i>et al.</i> (1978)	2.2	6.7
Antunes <i>et al.</i> (1979)	6.0	15.0
Hulka <i>et al.</i> (1980)	—	3.6
Shapiro <i>et al.</i> (1980)	3.9	6.0
Jelovsek <i>et al.</i> (1980)	2.4	4.8
Spengler <i>et al.</i> (1981)	3.2	8.6
Stavvaky <i>et al.</i> (1981)	4.2	14.4
Kelsey <i>et al.</i> (1982)	—	8.2

^aRisk relative to never-users.

Thus the evidence for a causal association between estrogen therapy and endometrial cancer is strong: the relative risks are high and the results have been confirmed by many studies, using different methodologies. In addition, dose response has been well documented. Importantly, the association is biologically plausible since it is known that unopposed estrogens stimulate hyperplastic changes in the endometrium (Whitehead *et al.*, 1979).

In this chapter, we are presenting preliminary data from yet another case-control study of endometrial cancer and estrogen therapy. As in all the other studies, we also have found an increased risk with use of estrogens.

This analysis was limited to the 136 women with endometrial cancer and the 558 control women enrolled in the Cancer and Steroid Hormone Study who had undergone a natural menopause, since women who had had a hysterectomy were not at risk for developing endometrial cancer. We excluded four cases and 20 controls who had a past history of use of sequential OCs, since these are also reported to be associated with an increase risk of endometrial cancer. In the logistic regression model, we controlled for the potentially confounding variables of age, parity, obesity, and OC use.

Women with endometrial cancer were more likely to be nulliparous (32% vs. 11%), obese (29% vs. 13%), and hypertensive (36% vs. 22%) than were control women. Also, cases were less likely to have ever used OCs (30% vs. 42%), which is not unexpected since OC use has been shown to protect women from endometrial cancer.

The number of cases and controls who had ever used estrogens was small, but note that women who had ever used menopausal estrogens had a relative risk of 3.2 in comparison with women who had never used estrogens (Table VIII). A trend of increasing risk was associated with increasing duration of estrogen use. For the few women who had used estrogens for 6 years or more, the risk of endometrial cancer was seven times greater than for women who had

Table VIII
Risk of Endometrial Cancer by Duration of Postmenopausal Estrogen Use^a

Duration of estrogen use ^b	Cases	Controls	Relative risk ^c (95% confidence interval)
Never	104	493	1 (referent)
Ever	26	41	3.2 (1.8-5.8)
< 2 years	8	19	2.1 (0.9-5.3)
2-5 years	9	15	2.8 (1.1-7.1)
≥ 6 years	9	7	7.2 (2.5-21.0)

^aFrom Centers for Disease Control Cancer and Steroid Hormone Study.

^bExcludes six cases and 34 controls with unknown duration of use.

^cLogistic regression estimate, adjusted for age, parity, obesity, and history of oral contraceptive use.

Table IX

Risk of Endometrial Cancer by Use of Postmenopausal Estrogens and by Past History of Oral Contraceptive Use^a

Duration of estrogen use ^c	Use of oral contraceptives ^b			
	Never		Ever	
	Cases/controls	Relative risk ^d	Cases/controls	Relative risk ^d
Never	74/298	1 (referent)	30/195	0.8
Ever	20/20	3.8 ^e	6/21	1.9
<2 years	5/10	1.7	3/9	2.5
2-5 years	7/7	3.7 ^e	2/8	1.3
≥6 years	8/3	11.5 ^e	1/4	2.1

^a From Centers for Disease Control Cancer and Steroid Hormone Study.

^b Excludes one case and two controls with unknown OC use.

^c Excludes six cases and 34 controls with unknown duration of estrogen use.

^d Logistic regression relative risk, adjusted for age, parity, and obesity.

^e $P < 0.05$.

never used estrogens. We also found a trend of increasing risk of endometrial cancer with increasing time since first use of estrogens.

Because epidemiologic evidence suggests that use of combination OCs prevents a woman from subsequently developing endometrial cancer, we examined the data to see if women who used OCs in addition to estrogens had a different risk of endometrial cancer than women who had used estrogens but had no history of OC use (Table IX). Compared with women who had never used estrogens or OCs, those who had used estrogens and OCs had an overall risk of 1.9 whereas women who had used estrogens but had no history of OC use had an overall risk of 3.8. We had few cases and controls in each duration category; however, the highest risk estimates were seen among the women who had longer duration of estrogen use, but no history of past OC use.

Insufficient numbers of past users existed to examine the risks after discontinuation of estrogen therapy. Likewise, few women in the study had used concurrent progestin therapy, so we were unable to determine whether this altered the risk estimates.

The results from this study support the results obtained in other case-control studies of estrogen therapy and endometrial cancer, both in the magnitude of the relative risks and the trend toward increasing risk with increasing duration of therapy. To date, no published study has had results suggesting that women with a history of both OC use and estrogen therapy have a different risk of endometrial cancer than do women who have used estrogens but never used OCs. We emphasize that this particular finding is not conclusive, but rather suggestive. Some previously published studies of estrogen therapy and endo-

metrial cancer have collected information on use of OCs; reanalysis of their data may clarify this question.

Conclusion

In summary, the preliminary results from the Cancer and Steroid Hormone Study are generally similar to the findings of previous studies concerning the association of estrogen therapy to breast, ovarian, and endometrial cancer. Our findings suggest that long-term estrogen users may have an increased risk of breast and ovarian cancer, although the risk elevations may well be due to chance or some unrecognized bias. Also, our results support the well-established findings that estrogen therapy is a risk factor for endometrial cancer. We have reported for the first time results that suggest that past OC use may modify the association between estrogen therapy and endometrial and ovarian cancers.

Studies are needed to definitively characterize the risks of cancer associated with estrogen therapy. For endometrial cancer, convincing information is needed to demonstrate that combined estrogen-progestin therapy does in fact reduce the increased risk of endometrial cancer associated with unopposed estrogen therapy. If it does, what are the long-term sequelae of combined therapy on other systems, particularly the cardiovascular system? Also, does past OC use really modify the association between endometrial cancer and estrogen therapy?

More puzzling questions surround breast and ovarian cancer: Does estrogen therapy increase a woman's risk of developing breast or ovarian cancer? How might combined estrogen and progestin therapy affect the risk of developing these two cancers? Obtaining answers to these questions will require specially designed studies with sufficient numbers of study subjects to detect differences of the magnitude demonstrated today.

If long-term estrogen therapy, with or without concomitant use of a progestin, is promoted to prevent osteoporosis and its serious sequelae, then we need to have a clear understanding of the association of long-term therapy with cancer. More information about the effects of estrogen therapy on different organ systems is needed so that the benefits and risks of estrogen therapy can be placed in perspective.

References

- Annegers, J. F., Strom, H., Decker, D. G., Dockerty, M. B., and O'Fallon, W. M., 1979, Ovarian cancer: Incidence and case-control study, *Cancer* **43**:723-729.
- Antunes, C. M. F., Stolley, P. D., Rosenshein, N. B., Davies, J. L., Tonascia, J. A., Brown, C., Burnett, L., Rutledge, A., Pokempner, M., and Garcia, R., 1979, Endometrial cancer and estrogen use: Report of a large case-control study, *N. Engl. J. Med.* **300**:9-13.

- Boston Collaborative Drug Surveillance Program, 1974, Surgically confirmed gallbladder disease, venous thromboembolism, and breast tumors in relation to postmenopausal estrogen therapy, *N. Engl. J. Med.* **290**:15-18.
- Brinton, L. A., Hoover, R. N., Szklo, M., and Fraumeni, J. F., 1981, Menopausal estrogen use and risk of breast cancer, *Cancer* **47**:2517-2522.
- Casagrande, J., Gerkins, V., Henderson, B. E., Mack, T., and Pike, M. C., 1976, Brief communication: Exogenous estrogens and breast cancer in women with natural menopause, *J. Natl. Cancer Inst.* **56**:839-841.
- Centers for Disease Control Cancer and Steroid Hormone Study, 1983a, Long-term oral contraceptive use and the risk of breast cancer, *JAMA* **249**:1591-1595.
- Centers for Disease Control Cancer and Steroid Hormone Study, 1983b, Oral contraceptive use and the risk of ovarian cancer, *JAMA* **249**:1596-1599.
- Centers for Disease Control Cancer and Steroid Hormone Study, 1983c, Oral contraceptive use and the risk of endometrial cancer, *JAMA* **249**:1600-1604.
- Craig, T. J., Comstock, G. W., and Geiser, P. B., 1974, Epidemiologic comparison of breast cancer patients with early and late onset of malignancy and general population controls, *J. Natl. Cancer Inst.* **53**:1577-1581.
- Cramer, D. W., Hutchinson, G. B., Welch, W. R., Scully, R. E., and Ryan, K. J., 1983, Determinants of ovarian cancer risk: I. Reproductive experiences and family history, *J. Natl. Cancer Inst.* **71**:711-716.
- Gray, L. A., Christopherson, W. M., and Hoover, R. N., 1977, Estrogens and endometrial carcinoma, *Obstet. Gynecol.* **49**:385-389.
- Hildreth, N. G., Kelsey, J. L., LiVolsi, V. A., Fischer, D. B., Holford, T. R., Mostow, E. D., Schwartz, P. E., and White, C., 1981, An epidemiologic study of epithelial carcinoma of the ovary, *Am. J. Epidemiol.* **114**:398-405.
- Hoogerland, D. L., Buchler, D. A., Crowley, J. J., and Carr, W. F., 1978, Estrogen use — risk of endometrial carcinoma, *Gynecol. Oncol.* **6**:451-458.
- Hoover, R., Gray, L. A., Sr, Cole, P., and MacMahon, B., 1976, Menopausal estrogens and breast cancer, *N. Engl. J. Med.* **295**:401-405.
- Hoover, R., Gray, L. A., Sr, and Fraumeni, J. F., Jr, 1977, Stilboestrol (Diethylstilbestrol) and the risk of ovarian cancer, *Lancet* **2**:533-534.
- Hoover, R., Glass, A., and Finkle, W. D., et al., 1981, Conjugated estrogens and breast cancer risk in women, *J. Natl. Cancer Inst.* **67**:815-820.
- Horwitz, R. I., and Feinstein, A. R., 1978, Alternative analytic methods for the case-control studies of estrogens and endometrial cancer, *N. Engl. J. Med.* **299**:1089-1094.
- Hulka, B. S., 1980, Effect of exogenous estrogen on postmenopausal women: The epidemiologic evidence, *Obstet. Gynecol. Surv.* **35**:389-399.
- Hulka, B. S., Grimson, R. C., Greenberg, B. G., Kaufman, D. G., Fowler, W. C., Hogue, C. J. R., Berger, G. S., and Pulliam, C. C., 1980, "Alternative" controls in a case-control study of endometrial cancer and exogenous estrogen, *Am. J. Epidemiol.* **112**:376-387.
- Hulka, B. S., Chambless, L. E., Deubner, D. C., and Wilkinson, W. E., 1982, Breast cancer and estrogen replacement therapy, *Am. J. Obstet. Gynecol.* **143**:638-644.
- Jelovsek, F. R., Hammond, C. B., Woodard, B. H., Draffin, R., Lee, K. L., Creasman, W. T., and Parker, R. T., 1980, Risk of exogenous estrogen therapy and endometrial cancer, *Am. J. Obstet. Gynecol.* **137**:85-91.
- Jick, H., Walker, A. M., Watkins, R. N., D'Ewart, D. C., Hunter, J. R., Danford, A., Madsen, S., Dinan, B. J., and Rothman, K. J., 1980, Replacement estrogens and breast cancer, *Am. J. Epidemiol.* **112**:586-594.
- Kaufman, D. W., Miller, D. R., Rosenberg, L., Helmrigh, S. P., Stolley, P., Schottenfeld, D., and Shapiro, S., 1984, Noncontraceptive estrogen use and the risk of breast cancer, *JAMA* **252**:63-67.

- Kelsey, J. L., Fischer, D. B., Holford, T. R., LiVolsi, V., Mostow, E. D., Goldenberg, I. S., and White, C., 1981, Exogenous estrogens and other factors in the epidemiology of breast cancer, *J. Natl. Cancer Inst.* **67**:327-333.
- Kelsey, J. L., LiVolsi, V. A., Holford, T. R., Fischer, D. B., Mostow, E. D., Schwartz, P. E., O'Connor, T., and White, C., 1982, A case-control study of cancer of the endometrium, *Am. J. Epidemiol.* **116**:333-342.
- LaVecchia, C., Liberati, A., and Franceschi, S., 1982, Noncontraceptive estrogen use and the occurrence of ovarian cancer, *J. Natl. Cancer Inst.* **69**:1207.
- Mack, T. M., Pike, M. C., Henderson, B. E., Pfeffer, R. I., Gerkins, V. R., Arthur, M., and Brown, S. E., 1976, Estrogens and endometrial cancer in a retirement community, *N. Engl. J. Med.* **294**:1262-1267.
- McDonald, T. W., Annegers, J. F., O'Fallan, W. M., Dockerty, M. B., Malkasian, G. D., and Kurland, L. T., 1977, Exogenous estrogen and endometrial carcinoma: Case-control and incidence study, *Am. J. Obstet. Gynecol.* **127**:572-580.
- McGowan, L., Parent, L., Lednar, W., and Norris, H. J., 1979, The woman at risk for developing ovarian cancer, *Gynecol. Oncol.* **7**:325-344.
- Paterson, M. E. L., Wade-Evans, T., Sturdee, D. W., Thom, M. H., and Studd, J. W. W., 1980, Endometrial disease after treatment with oestrogens and progestogens in the climacteric, *Br. Med. J.* **280**:822-824.
- Ravnihar, B., Seigel, D. G., and Lindtner, J., 1979, An epidemiologic study of breast cancer and benign breast neoplasias in relation to the oral contraceptive and estrogen use, *Europ. J. Cancer* **15**:395-405.
- Ross, R. K., Paganini-Hill, A., Gerkins, V. R., Mack, T. M., Pfeffer, R., Arthur, M., and Henderson, B. E., 1980, A case-control study of menopausal estrogen therapy and breast cancer, *JAMA* **243**:1635-1639.
- Sartwell, P. E., Arthes, F. G., and Tonascia, J. A., 1977, Exogenous hormones, reproductive history and breast cancer, *J. Natl. Cancer Inst.* **59**:1589-1592.
- Shapiro, S., Kaufman, D. W., Slone, D., Rosenberg, L., Miettinen, O. S., Stolley, P. D., Rosenshein, N. B., Watring, W. G., Leavitt, T., and Knapp, R. C., 1980, Recent and past use of conjugated estrogens in relation to adenocarcinoma of the endometrium, *N. Engl. J. Med.* **303**:485-489.
- Silverberg, E. and Lubera, J. A., 1983, A review of American Cancer Society estimates of cancer cases and deaths, *CA* **33**:2-25.
- Smith, D. C., Prentice, R., Thompson, D. J., and Herrmann, W. L., 1975, Association of exogenous estrogen and endometrial carcinoma, *N. Engl. J. Med.* **293**:1164-1167.
- Spengler, R. F., Clarke, E. A., Woolever, C. A., Newman, A. M., and Osborn, R. W., 1981, Exogenous estrogens and endometrial cancer: A case-control study and assessment of potential biases, *Am. J. Epidemiol.* **114**:497-506.
- Stadel, B. V., 1975, The etiology and prevention of ovarian cancer, *Am. J. Obstet. Gynecol.* **123**:772-773.
- Stavraky, K. M., Collins, J. A., Donner, A., and Wells, G. A., 1981, A comparison of estrogen use by women with endometrial cancer, gynecologic disorders and other illnesses, *Am. J. Obstet. Gynecol.* **141**:547-555.
- Symmers, W. S., 1968, Carcinoma of the breast in transsexual individuals after surgical and hormonal interference with the primary and secondary sex characteristics, *Br. Med. J.* **2**:82.
- Thomas, D. B., 1978, Role of exogenous female hormone in altering the risk of benign and malignant neoplasms in humans, *Cancer Res.* **38**:3991-4000.
- Thomas, D. B., 1984, Do hormones cause breast cancer? *Cancer* **53**:595-604.
- Waksberg, J., 1978, Sampling methods for random digit dialing, *J. Am. Stat. Assoc.* **73**:40-46.

- Weiss, N. S., Lyon, J. L., Krishnamurthy, S., Diftert, S. E., Liff, J. M., and Daling, J. R., 1982, Noncontraceptive estrogen use and the occurrence of ovarian cancer, *J. Natl. Cancer Inst.* **68**:95-98.
- Whitehead, M. I., King, R. J. B., McQueen, J., and Cambell, S., 1979, Endometrial histology and biochemistry in climacteric women during oestrogen and oestrogen/progestogen therapy, *J. R. Soc. Med.* **72**:322-327
- Ziel, H. K., and Finkel, W. D., 1975, Increased risk of endometrial carcinoma among users of conjugated estrogens, *N. Engl. J. Med.* **293**:1167-1170.